Name of the module: **bitRev1024sEbyE**
Target delivery date: 04-17-17
Location in GIT repository: vspa_common\vspa-lib\bitRev\ bitRev1024sEbyE

Type of code:

IPPU assembly function (1 function)
C function invoking IPPU (1 function)

IPPU Function API:

```
extern void bitRev1024sEbyE(void);
```

C function invoking IPPU function API:

```
extern bool bitRev1024sEbyEInvoke(
    vspa_complex_float16        *pOut,
    vspa_complex_float16 const  *pIn
);
```

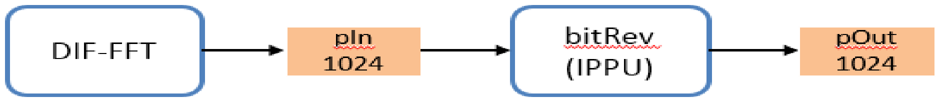
Return value: True when IPPU is not busy, false when IPPU is busy

API description:

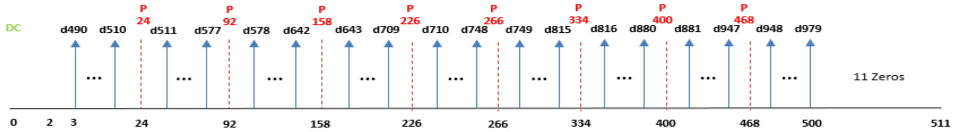
bitRev module performs shifting and bit-reversal on the input buffer of 1024 sub-carriers.

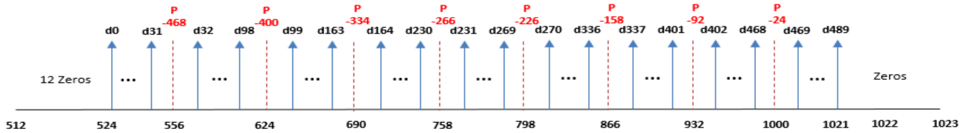
- pOut — pointer to linear ordered sub-carriers buffer in VCPU. (32 bit aligned)
- pIn — pointer to the input buffer in VCPU dmem containing sub-carriers in bit-reversed order. (dmem aligned)

Implementation Plan:

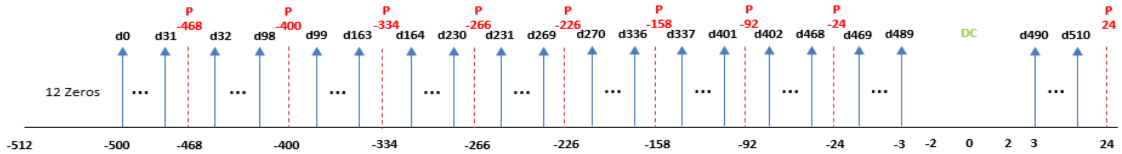


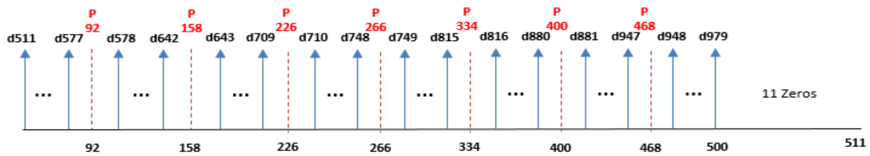
Here is the sub-carrier arrangement in input buffer WITHOUT bit-reversal addressing(Note: the module expects the following sub-carriers in bit-reversed address)





For the above input(in bit-reversed order) the output looks like the following





DMEM requirements:

Variable	Minimum size (half-words)	Minimum alignment (half-words)	Allocated by
pOut	2048	2	Caller
pIn	2048	64	Caller

Test Plan:

Verify the bitRev output with matlab reference for the following use-cases

- 802.11ax 80 MHz with 996-RU